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Responsible use of AI in social science research

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ABSTRACT

The paper elaborates on the responsible use of AI in social science research. It outlines the principles of responsible AI use in research, namely: transparency, traceability/auditability, explainability, fairness, privacy and data protection, respect and awareness of the broader AI impacts, human oversight and quality control, and accountability. It delves into the responsible AI use in each step of the research process from ideation to manuscript evaluation. The paper formulates specific recommendations for authors, PhD students, early career researchers and PhD programmes, publishers, funders, and policymakers related to responsible AI use in research. Finally, the paper sketches the future prospects of AI use in research and future research directions.

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Artificial intelligence;
research; social science;
responsible AI; research
ethics

1. Introduction

Artificial intelligence (AI) has long been used in social science research, usually for data analysis and proofreading (Chaudhary & Alam, 2024; Hemachandran et al., 2023). Until the launch of ChatGPT in November 2022, AI application in social science research required substantial technical skills limiting the number of social scientists who used it. The advancements in generative AI (GenAI) democratised AI. They allowed more researchers to use it as a tool in their studies and widened the scope of tasks in all stages of the research process AI could be utilised for (Bail, 2024; Davidson, 2024; Dwivedi et al., 2023; Ivanov & Soliman, 2023). Researchers and institutions were unprepared for the perfect storm induced by GenAI as it decreased the costs of unethical research practices and made the latter more accessible. GenAI facilitated the quick and cheap fabrication of content (text, tables, figures) and references in a convincing way (Walters & Wilder, 2023). Paper mill companies adopted GenAI to quickly fabricate whole papers that had already found their way into the academic record, sometimes leading to retractions (Chauhan & Currie, 2024; Nazarovets, 2024). AI-generated commentaries and short papers started to pollute academic journals and distort journal metrics (Joelving, 2024). Universities and research organisations had to rewrite their internal regulations and ethics guidelines, sometimes in a trial-and-error fashion (Driessens & Pischetola, 2024). Many publishers initially forbade the use of AI in scholarly papers besides proofreading,

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but later, they were forced to accept the new technological realities and started to request author declarations about the use of AI in manuscript preparation.

Many studies have focused on the implications of AI for research (Bagenal, 2024; Dwivedi et al., 2023), including in service research (Do, 2025; Imschloss et al., 2025). The literature discusses the research tasks AI can be used for (Marshall & Naff, 2024), the acceptance of GenAI by researchers (Ivanov et al., 2024; Strzelecki et al., 2024; Zou & Huang, 2023), the infiltration of AI-generated content into the scholarly record (Strzelecki, 2025), development of AI agents as research assistants (Schmidgall et al., 2025), reviewers' perceptions of GenAI in academic writing (Hadan et al., 2024), AI authorship of publications (Sharifzadeh, 2024), and ethical aspects of AI use (Andrade-Hidalgo et al., 2024; Brown & Heitner, 2024; Gabel et al., 2023; Marshall & Naff, 2024), among other topics. In general, authors agree that AI is revolutionising every stage of the research process and is making researchers more efficient (Andersen et al., 2025; Tingelhoff et al., 2024). Although studies point out plagiarism, content fabrication and other ethical risks posed by GenAI (Farangi et al., 2024; Ivanov & Soliman, 2023), they nonetheless acknowledge that AI can be a powerful ally to social scientists if used responsibly. However, studies usually elaborate on the responsible AI use in research on a highly theoretical level which is not always operationalised into specific actions that research stakeholders (i.e. authors, reviewers, editors, publishers, funders, policymakers, etc.) need to follow to ensure that they utilise AI responsibly, forming a gap in the literature. Moreover, recent studies have pointed out that researchers were split regarding AI ethics in research and have quite diverse opinions about the acceptable use of AI for academic writing or peer review (Kwon, 2025a). Therefore, this paper tries to fill this gap in the literature and aims to outline and critically evaluate potential (mis)uses of AI in the research process and elaborate practical recommendations for the responsible use of AI in social science research. Similar to previous studies (e.g. Iskender, 2023; Soliman et al., 2023; among many others), the paper illustrates the potential of AI with two conversations with GenAI applications (ChatGPT 4o and Claude 3.5 Sonnet) structured around the stages of the research process, which are included in Appendix 1.

The rest of the paper is organised as follows. The next section provides a focused literature review on the research process, research ethics and the principles of responsible AI use in research. Section 3 delves into the responsible AI use in each step of the research process from ideation to manuscript evaluation. Section 4 summarises the contributions of the paper, discusses the implications for authors, PhD students, early career researchers and PhD programmes, publishers, funders, and policymakers, sketches the future prospects of AI use in social science research, outlines the limitations and future research directions, and concludes the paper.

2. Literature review

2.1. The research process

To assess the responsible use of AI in social science research, it is convenient to look at the research as a process that includes various steps that researchers have to follow to produce a high-quality piece of research (see also Marshall & Naff, 2024). Kothari (2004; p. 11) identifies 7 steps of the research process: define the research problem, review

the literature, formulate hypotheses, design research, collect data, analyse data, and interpret and report. Creswell (2012; p. 8) follows a similar line and distinguishes 6 steps: identifying a research problem, reviewing the literature, specifying a purpose for research, collecting data, analysing and interpreting data, and reporting and evaluating research. In their classic textbook on research methods for business students, Saunders et al. (2023) provide a more detailed perspective of the steps in the research process that includes: generating research ideas and developing the proposal, critically reviewing the literature, understanding the research philosophy and approach, formulating research design, negotiating access and address ethical issues, planning data collection and collecting data, analysing data, and writing the project. It is evident that researchers include more or less the same steps from problem formulation to writing the project/manuscript. In practice, the research process starts with ideation when researchers generate ideas about what to research. The ideation results in a broad idea about the topic which is later narrowed down to a specific research problem. It is often considered a part of the problem definition (e.g. Saunders et al., 2023) but it can also be perceived as a separate step in the research process that demands inspiration and creativity. Once prepared, a manuscript goes through peer review which most often leads to additional reading and critical evaluation of literature, additional analyses, revising texts in the manuscript, etc. Hence, the research process needs to end up with the manuscript evaluation. Figure 1 outlines the steps in the research process used in this paper to evaluate the responsible use of AI in research, namely: ideation, problem definition, review of previous literature, formulation of research design, data collection, data analysis, manuscript preparation and manuscript evaluation.

2.2. Ethics in social science research

It is undeniable that research needs to be performed ethically (Creswell, 2012; Israel, 2015; Saunders et al., 2023). The research process and its output should not put anyone in harm – the research participants and the organisations they represent, the researchers and their institutions, the managers and policymakers who will use the findings of the research, and any third party. Good research practices require that respondents provide explicit informed consent to participate voluntarily in the research and have the right to withdraw at any time without providing a reason. Researchers are expected to develop research instruments that help collect valid data (i.e. the instrument measures what it is intended to measure), collect data from a sufficient number of relevant respondents, ensure the privacy, confidentiality and anonymity of respondents and their affiliations (as necessary), implement proper (statistical) analysis, ensure transparency in and correct reporting of results, write original texts, etc. Practices such as plagiarism, biased samples and datasets, data falsification and fabrication, improper coding of raw data, cherry-picking of results, changing/inventing quotes from previous publications and respondents, lack of methodological transparency, fake reviews and references, incorrect use of terms and many others violate good ethical practices, lead to studies that are not valid or reliable and misinform other studies, business decisions and policy-making (Ben-Yehuda & Oliver-Lumerman, 2017; D'Angelo, 2019; Wible, 2023). If unethical practices are identified in already published papers, they are usually retracted (Alrawadieh & Zareer, 2019). Every university and research organisation has ethical guidelines for all steps of the research process and

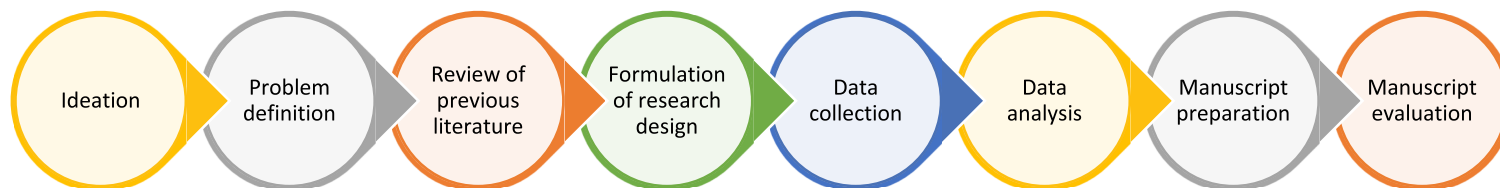


Figure 1. Steps in the research process.

organises ethics training for its (doctoral) students and staff (Altinay et al., 2016). Ethical guidelines also regulate research data management and data access. Many institutions have an ethics review board and require students and staff to have an approved ethics application before starting data collection. The ethics approval process aims to prevent ethical issues before they arise rather than punishing researchers.

2.3. Principles of responsible AI use in research

One of the major directions of research on the intersection between AI and research ethics is the principles of responsible use of AI in research that shape the specific practices to be adopted by researchers, authors, reviewers, editors, publishers, universities and everyone else who benefits from or is affected by AI in research. The International Organization for Standardization (ISO, n.d.) stipulates that responsible AI should be based on the principles of fairness, transparency, non-maleficence, accountability, privacy, robustness, and inclusiveness. OECD (2024) formulates five value-based principles: inclusive growth, sustainable development and well-being; human rights and democratic values, including fairness and privacy; transparency and explainability; robustness, security and safety; and accountability. It is evident that both ISO's and OECD's principles are largely overlapping and focus mostly on the development and use of AI by organisations and policymakers. In research, the European Commission (2024) in its *Living guidelines on the responsible use of generative AI in research* formulates four basic principles: reliability, honesty, respect and accountability. In their scoping review, Bouhouita-Guermech et al. (2023) derive the following principles and norms of AI in research ethics: the application of ethical approaches in terms of AI research ethics; responsibility; explainability and validity; transparency and informed consent; privacy; data protection; and privacy vs. science advancement dilemmas. Knoechel et al. (2024) consider regulations, data security, quality control, originality, bias mitigation, accountability, transparency, and awareness of the broader impacts of AI use. Stepping on these frameworks, we can summarise the following principles of responsible AI use in social sciences:

- *Transparency* – complete disclosure of all aspects of AI use – software (including version), prompts (if GenAI), tasks for which AI was used, date/period of use (if relevant, especially in GenAI), the person(s) who used it, datasets, etc. (Resnik & Hosseini, 2024). Transparency requires researchers to elaborate on all administrative and operational details of AI use to ensure that the interested stakeholders (editors, publishers, readers, funders, research participants, etc.) have sufficient information about the AI use in a particular research process and its outputs. This is a principle frequently broken, as thousands of academic papers are suspected to have undisclosed AI content (see Kwon, 2025b).
- *Traceability/Auditability* – all steps of AI use are documented, reported, can be followed and verified (Li & Goel, 2024). This allows interested stakeholders to audit/check the training datasets, prompts and outputs and ensure the reproducibility of the study and the validity and reliability of the results. For GenAI, traceability/auditability is achieved through the provision of the link to the conversation with the respective AI application so readers can audit it.
- *Explainability* – AI users need to understand and interpret AI decisions and output and how the AI reached them (Gunning et al., 2019). This is of major concern because often

AI applications act as black boxes and for their users, it is difficult to understand how AI reached particular conclusions/decisions. Moreover, explainability needs to address the information needs of people who do not necessarily have a background in computer sciences and have a limited understanding of how AI works but use it. Therefore, explanations must be in plain language (Resnik & Hosseini, 2024). In GenAI tools, users may ask for justification and explanation of the generated output (text, video, sound, photo, code).

- *Fairness* relates to the mitigation of potential biases in AI use such as biases in the training dataset (e.g. overrepresentation of data from some social/respondent groups and underrepresentation of others) or the algorithms (the AI decisions put some groups in a disadvantaged position), among others (Giovanola & Tiribelli, 2023; Ivanov & Umbrello, 2021). Biases in the training datasets and algorithms are especially concerning because they may lead to wrong conclusions in the research process even if the analysed data is reliable. For example, the AI application may misinterpret the tone and meaning of analysed text or generate content (e.g. text or photo) that is insulting to or misrepresenting specific social groups.
- *Privacy and data protection* – the use of anonymised data in AI tools and compliance with relevant data protection regulations (e.g. GDPR).
- *Respect* for all research stakeholders (other researchers, respondents, third parties, societies, etc.) that might be affected by the [improper] use of AI (European Commission, 2024) and awareness of the broader societal, economic, political and environmental impacts of AI use in research.
- *Human oversight and quality control* of the AI's implementation in the research process and AI's output (Knoechel et al., 2024). This requires keeping the human 'in the loop' (AI recommends, the human decides) or 'on the loop' (AI decides and implements, the human can override and change the decision) rather than outsourcing the key research decisions to AI. As different organisations may have different rules, collaboration between researchers from different institutions may require a common agreed upon oversight procedures.
- *Accountability* relates to the clear roles in AI use in the research team members and taking responsibility for all outputs produced with the help of AI (European Commission, 2024). This requires the researchers to report explicitly who was using AI in a particular study and how, and to take full responsibility in case of improper use of AI and its outputs.

Adherence to the above principles does not guarantee that AI-assisted research will be of high-quality but that it will be implemented in an ethical way. In practice, however, the principles are sometimes contradictory and adhering strictly to one may mean significant compromises with another. For example, the principles of transparency and traceability requires that authors share all the details related to AI use. While some of them such as used software and its version, prompts, anonymised datasets, etc., can be easily shared, the intellectual property rights put limitations on the degree of sharing information about the algorithms. While authors can mention in their papers that they used ChatGPT 4o, they cannot provide any information about its algorithms because they are proprietary and largely remain a black box for the AI tool users. Hence, the public's interest in transparency of the research and AI use goes in conflict with IT companies'

intellectual property rights that need to protect their financial investments in the development of the respective AI algorithms.

3. Responsible use of AI in the social science research process

3.1. Ideation

Large Language Models (LLMs) are an excellent source of ideas for research projects. For transparency and traceability, Appendix 1 shows the author's conversations with two popular LLM chatbot applications, including the respective links to them. The LLM chatbots were asked to generate ideas about a potential research topic in the field of service robots. The field was selected due to its specificity, novelty, and an author's familiarity with it. Both applications proposed numerous ideas (ChatGPT – 16, Claude – 30) that ranged from technical aspects of robots, social and ethical issues to robots' application in particular contexts. A comparison of the ideas shows that many of them overlap, e.g. human-robot interaction, task management, learning capabilities of robots, robustness to operate in diverse environments, ethics, etc. This is not surprising, considering that both applications have been trained with similar datasets of academic papers. A closer look at the proposed topics shows that all of them are relevant and feasible. However, there are several potential pitfalls for researchers:

- The proposed topics are more or less similar in both applications. Therefore, authors who rely on generative AI for research ideas may end up researching similar things. Although this is not a problem per se, authors will lose their competitive advantage if many research teams submit papers on virtually the same topic.
- The proposed topics are indeed important but they have been extensively researched. This is not surprising, considering the way LLMs work – the more frequent a topic is in the training article dataset, the more likely it is to be produced as an output to a prompt. But an extensively researched topic is not necessarily an important topic; just the opposite. In this particular example, the novelty of the proposed topics is marginal. The LLMs did not mention anything related to the economics of service robots, their impacts on tasks and jobs, their integration with other existing technologies in companies, the societal and macroeconomic effects of service robots, the design of robot-friendly facilities or other topics that form a gap in the literature. Hence, the LLMs formulated good topics in areas we already know much about but none in areas that we know little about and need research.

Therefore, human discretion and proper prompt engineering are needed to filter out unattractive topics and narrow the proposals down to a topic that has significant potential for development. Of course, this requires prior reading by the author rather than over-reliance on AI.

3.2. Problem definition

AI works well in specific contexts. Once the research topic has been identified, AI can help in formulating the research problem, aim, and objectives. Appendix 1 shows the problem definition, research aim and objectives generated by ChatGPT and Claude for one of the

topics that was initially proposed by ChatGPT. As evident, the texts are well formulated, although some human touch would be necessary. The comparison of the outputs of the two applications shows that the aim is nearly identically formulated with only minor differences in wording. The objectives are also relevant and largely overlap between the two applications. The research problem definition provided by the apps can serve as an excellent backbone of the future paper's rationale after some expansion, including with the help of AI. The simultaneous use of several AI applications can help researchers in identifying and formulating the aim and objectives by adapting the most appropriate and relevant texts by the apps.

The responsible use of AI at this stage requires significant attention in utilising AI-generated texts. The aim and objectives need to be coherent but also achievable within the respective publication. A look at the proposed aim and objectives in Appendix 1 reveals that the objectives proposed by the AI apps are very ambitious for a journal article but suitable for a PhD thesis, a research monograph, or a series of articles/conference papers. In practice, each proposed objective can serve as an aim for a separate publication.

3.3. Review of previous studies

Although GenAI can effectively assist in preparing literature reviews (Bolanos et al., 2024; Tingelhoff et al., 2024), this is one of the areas that raised significant ethical concerns among researchers and publishers giving ground to some researchers not to recommend the use of AI for literature reviews (see Cheng et al., 2025). When it was launched, ChatGPT invented references, a problem known as 'hallucination' (Ivanov & Soliman, 2023). Unfortunately, many researchers violated the human oversight and AI quality control principle and blindly cited the fabricated references without checking whether they existed. While some fictitious references were caught during the editing and copyediting stage, others were left and are now part of the scholarly record. This is a major publication ethics issue because the authors refer to sources that do not exist. For traceability, some publishers reacted by requesting DOI or other links to the actual publication as a guarantee that the paper existed. The responsible use of AI for literature review requires an explicit and thorough check of every single reference provided by the respective AI application to ensure its existence and relevance to the topic (human oversight and AI quality control principle). The AI output in Appendix 1 shows that ChatGPT provided references to every paragraph it wrote (the doi links in the Appendix were added by the author of this paper due to the inability to copy/paste the text from ChatGPT with the references). However, the references were only two; hence, the review was quite basic. Nevertheless, it could serve as a good draft to be expended by the researchers. Similarly, Perplexity.ai (<https://www.perplexity.ai/search/please-provide-a-review-of-aca-iYYooe9mRNKNmzQJy3NrRw#0>) provided links to references at the end of each paragraph while Scite.ai included them in each sentence and provided snippets from the sources to show the citation context (see Figure 2a). At the same time, Claude produced a very relevant text but without links to the sources. When asked to provide the links to the cited references, Claude answered that it could not do it. Hence, in practice, the text by Claude is unusable because researchers cannot know which sources were cited and used to generate it (i.e. the transparency and traceability principles were violated).



● Please provide a review of academic publications on how cultural differences aff...

Export References



Cultural differences significantly influence human-robot interactions (HRI), affecting various dimensions such as acceptance, communication styles, and cognitive processes. A comprehensive review of the literature reveals that cultural backgrounds shape attitudes toward robots, nonverbal behaviors, and the overall effectiveness of HRI.

One of the primary areas of research highlights how cultural context impacts attitudes towards humanoid and animal-like robots. Papadopoulos and Koulouglioti [Papadopoulos & Koulouglioti \(2018\)](#) emphasize that cultural backgrounds dictate not only preferences but also the physical proximity maintained during interactions. For instance, Eresha et al. found that Arab participants tended to position themselves closer to humanoid robots compared to their German counterparts, indicating a cultural variance in comfort levels during HRI ([Papadopoulos & Koulouglioti, 2018](#)). This finding aligns with the broader understanding that nonverbal communication, which varies across cultures, plays a crucial role in shaping the dynamics of HRI ([Leoste, 2024](#)).

Moreover, the importance of mutual understanding in HRI is underscored by Sciutti et al., who argue that robots must be designed to interpret cultural signals accurately. They note that while some behaviors, such as eye gaze, may have universal interpretations, others, like hand gestures, require cultural context for proper understanding ([Sciutti et al., 2018](#)). This necessity for cultural sensitivity is further supported by Marchesi et al., who discuss how cultural values influence cognitive processes like anthropomorphism and mind attribution towards robots, suggesting that these processes are not universally applicable but rather culturally contingent ([Marchesi et al., 2021](#)).

The acceptance of robots is also heavily influenced by cultural factors. Bröhl et al. developed a Human-Robot Collaboration Acceptance Model that incorporates cultural norms, demonstrating that individuals are more likely to accept robots that align with their cultural expectations ([Bröhl et al., 2019](#)). Similarly, Lim et al. synthesized empirical evidence indicating that cultural influences shape expectations and responses to social robots, highlighting the intricate relationship between culture and human cognition in HRI contexts ([Lim et al., 2020](#)). This is echoed in the work of Korn et al., who found significant differences in preferences for robot attributes among participants from different cultural backgrounds, further illustrating the necessity of culturally aware robot design ([Korn et al., 2021](#)).

↳ Ask a question... (type "/" for menu)

References

Search Strategy



"...A person's cultural background was also found to affect nonverbal behaviors during human-robot interactions, as has been obse..." >>

Section: Results

The Influence of Culture on Attitudes Towards Humanoid and Animal-like Robots: An Integrative Review

Irena Papadopoulos¹, Christina Koulouglioti² 2018

J of Nursing Scholarship

61 1 47 0

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"...This study presents a literature review focused on nonverbal communication in human-robot interaction (HRI) that involves service robots with social capabilities. We aim to list the types of robots used and nonverbal communication cues examined in the reviewed studies; and the main research objectives, participant characteristics, data collection methods, and primary findings of these studies. To achieve this, we conducted a literature review of 39 relevant open access academic papers published from 2006 to 2023 that examined the utilization of nonverbal cues by both humans and robots during HRI..." >>

Section: Abstract

Nonverbal Behavior of Service Robots in Social Interactions – A Survey on Recent Studies

Janika Leoste 2024

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"...In this respect, roboticists will have to take into consideration



a) Review by Scite.ai

Figure 2. Review of academic articles.

Explore topics and discover relevant references since 2003 [How it works](#)

Please provide a review of academic publications on how cultural differences affect human-robot interactions



Please provide a review of academic publications on how cultural differences affect human-robot interactions

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Summary

Review of Academic Publications on Cultural Differences in Human-Robot Interactions

Cultural differences significantly impact human-robot interactions (HRI), influencing user expectations, perceptions, and behaviors. Here is a summary of key findings from recent academic publications:

Impact on Robot Behavior and Acceptance

- **Cultural Adaptation:** Robots that adapt their behavior to align with the cultural context of users are more likely to be accepted and integrated into daily life. For instance, robots can learn to behave socially in alignment with an individual's cultural background, generating different behavioral responses to the same stimuli based on cultural context [1](#).
- **Task Performance:** The cultural background of users affects their expectations of robot behavior, especially when the robot fails to complete a task and requires user support. This necessitates robots to adapt their behavior according to the user's cultural background [2](#).

Perception and Interaction Preferences

- **Robot Appearance and Task:** Cultural differences influence how users perceive the likeability, engagement, trust, and satisfaction with robots. For example, participants from different cultures (Chinese, Korean, and German) showed varying preferences for robot appearance and task performance [3](#).
- **Interpersonal Distance:** Cultural norms regarding personal space also apply to interactions with robots. For example, users from the U.S. and India exhibited different interpersonal distance behaviors when operating telepresence robots, reflecting their cultural preferences [4](#).

Cultural Dichotomies and Preferences

- **Individualism vs. Collectivism:** Implementing cultural dichotomies, such as individualism vs. collectivism, in robot behavior can influence user preferences. However, these manipulations may not always align with personal preferences, as seen in studies with German and Japanese participants [5](#).
- **Multi-Robot Systems:** Cultural differences also affect user preferences and perceived usefulness of multi-robot systems. Significant differences were found between participants from India, China, and the USA, highlighting the importance of considering cultural factors in the design of domestic multi-robotic assistants [6](#).

User Experience and Interaction Context

- **Health and Assistive Robots:** Cultural background affects how users interact with socially-assistive robots (SARs). For example, American users focused more on the interactive experience, while Korean users critiqued the technical aspects, viewing the robots more as tools [7](#).

References

Reference 1

Encoding cultures in robot emotion representation
Le Quyen Dong T., Tuyen N.T.V., Jeong S., Chong N.Y.
RO-MAN 2017 - 26th IEEE International Symposium on Robot and Human Interactive Communication ↗
2017

Reference 2

Investigating the influence of culture on helping behavior towards service robots
Bajones M., Weiss A., Vincze M.
ACM/IEEE International Conference on Human-Robot Interaction ↗
2017

Reference 3

A cross-cultural study: Effect of robot appearance and task
Rau P.L.P., Li Y., Li D.
International Journal of Social Robotics ↗ 2010

[Show all 10 references](#)

Foundational documents

169 citations

Effects of communication style and culture on ability to accept recommendations from robots
P.P., Rau, Pei-Luen Patrick, Y. Li, Ye, D., Li, Dingjun
Computers in Human Behavior ↗ 2009

135 citations

When in Rome: The role of culture & context in adherence to robot recommendations
L., Wang, Lin, P.P., Rau, Pei-Luen Patrick, V., Evers, Vanessa, ([...], P.J., Hinds, Pamela J.

[Show more documents](#)

b) Review by Scopus AI

Figure 2 Continued

Scopus AI, launched in 2024, allows a review of papers in Scopus (Figure 2b). Although these are existing papers (otherwise they would not be indexed in Scopus), their review is based only on their abstracts, not the full text. The inclusion/exclusion of papers is entirely grounded only on a small fraction of the papers' text (the title, abstract and keywords). Therefore, important and relevant publications may be overlooked and omitted in the review because the algorithms do not have access to their full text. Additionally, papers not indexed in Scopus are not included in its reviews. At the same time, some publishers launched their own chatbots but their training is limited to the full-text of the papers in their own journals/books. Hence, they may provide information from and cite predominantly (if not only) publications owned by the publisher. From an ethical perspective, although very efficient and transparent, the use of such research-focused AI chatbots trained on part of the text (title, abstract, key words) or a selection of papers (owned by the publisher only) may not adhere to the fairness principle of AI use due to the inherent bias in the training dataset and the exclusion of much relevant content in it. While this exclusion is inevitable and acceptable at the currently introductory stage of research-focused AI chatbots, their future versions need to overcome this issue.

One of the applications that can save researchers a lot of time is the summarisation of papers. Some AI assistants provide a summary of pdf files. Figure 3 shows the summary prepared by the AI Assistant embedded in Adobe Reader of one of the open-access sources cited by ChatGPT in its literature review and a chat with Nature Research Assistant about the same paper. AI helps to distil the core ideas of the paper, the key points in its methodology, and the main conclusions. For transparency and traceability, the generated highlights in Adobe Reader AI Assistant include also a link to the respective sentences in the main text where the issues are discussed. Such AI assistants allow researchers to read (or 'scan') many more publications than they actually can without the AI because the AI will emphasise the important aspects of the paper and those of interest to the researcher and mentioned in the respective prompt, and will filter out the less important one. In that way, researchers can have a more focused, productive and efficient reading of the academic literature.

However, reliance on AI in the literature review has many drawbacks (Ivanov, 2023). First, the generated texts tend to be very descriptive. They lack critical evaluation and are nothing more than a summary of the main points. An academic paper requires an in-depth critical evaluation of the literature which the AI, currently, cannot provide. Second, the cited publications may not always be relevant or carry sufficient academic weight. AI may cite what is available in its database (Scopus AI) or on the internet (ChatGPT, Perplexity, Scite.ai) but with little effort for the critical selection of the sources. However, as researchers know well, publications are not made equal and some are seminal and fundamental to a topic while others have marginal contribution or relevance. Third, generative AI may fabricate references or not provide any at all. In both cases, researchers will have ethical problems if they use the texts unedited. Fourth, some AI-generated reviews step only on the abstracts of papers, not the full text. Hence, they may not reflect fully what the paper says. Fifth, there might be substantial differences in what the AI and the researchers consider as important in a paper. The AI might overlook the small sample size or mistakes in the research design or the data collection instrument but these should be of vital importance for the researchers because they may invalidate the study's findings. The responsible use of AI for literature reviews

All tools Edit Convert E-Sign

Find text or tools

AI Assistant

Management and Organization Review (2023), 19, 859–875
doi:10.1017/mor.2023.21

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PERSPECTIVES

Cultural Differences in People's Reactions and Applications of Robots, Algorithms, and Artificial Intelligence

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Abstract

Although research in cultural psychology has established that virtually all human behaviors and cognitions are in some ways shaped by culture, culture has been surprisingly absent from the emerging literature on the psychology of technology. In this perspective article, we first review recent findings on machine aversion versus appreciation. We then offer a cross-cultural perspective in understanding how people might react differently to machines. We propose three frameworks – historical, religious, and exposure – to explain how Asians might be more accepting of machines than their Western counterparts. We end the article by discussing three exciting human–machine applications found primarily in Asia and provide future research directions.

摘要

文化心理学的研究表明, 几乎所有人类行为和认知都在某种程度上受到文化的影响。但令人惊讶的是, 在新兴的技术心理学文献中, 几乎没有对于文化这一因素的关。在这篇前瞻性文章中, 作者首先回顾了现有文献中关于厌恶机器算法和欣赏机器算法的研究成果, 发现亚洲人比西方人更接受机器算法。作者通过跨文化视角, 提出了三个框架——历史、宗教、曝光度, 来解释为什么亚洲人和西方人会对机器算法产生截然不同的反应。最后, 作者讨论了在亚洲兴起的三种主要的有趣的人机应用程序, 并预测了未来的研究方向。

Keywords: algorithms; artificial intelligence; culture; human–machine interaction; robots
关键词: 算法; 人工智能; 文化; 人-机互动; 机器人

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Overview

The document discusses cultural differences in people's reactions and applications of robots, algorithms, and artificial intelligence, with a focus on how Asians may be more accepting of machines than Western counterparts.

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- Cultural differences in attitudes toward machines, influenced by historical, religious, and exposure factors. ¹
- Growth of robots and AI in Asia, showcasing diverse applications like robot-manned hotels and caregivers. ²
- Concerns and benefits of emerging technologies like sex robots, robot priests, and robot therapists. ³
- Research areas covering cultural psychology, human-robot interaction, and ethical considerations. ⁴
- Future research directions emphasizing cultural differences in human-machine interactions and new technological applications in Asia. ⁵

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a) Adobe Reader AI Assistant

Figure 3. Summary of an academic article by an AI assistant.

b) Nature Research assistant

Figure 3 Continued

requires the active involvement of the researcher in designing appropriate prompts to deliver targeted results and an explicit check of the full text of all cited references. The generated texts need to be edited because they can serve only as a starting point for the critical evaluation of the literature. A good practice is the creation of own private customised GPT (generative pre-trained transformer) on ChatGPT or elsewhere with the pdf files of selected articles or their simultaneous analysis with various AI applications (e.g. Perplexity, or AI Assistant for Adobe Reader), provided the uploading of files does not violate any copyrights. In that way, the researcher keeps control of the input of academic publications in the AI's database that will be used to prepare the review (the human oversight principle). Finally, the overreliance on AI for literature reviews may lead to loss of reading and critical thinking/evaluation skills in researchers. Studies have already shown that overreliance on AI chatbots decreases brain activity (Kosmyna et al., 2025). Prioritising the quantity of publications over their quality in performance appraisals of researchers inevitably pushes them into finding ways to increase their publication output and utilising AI to speedup reading and writing at the expense of the time for critical thinking and reflection. While in the short run such a strategy might contribute to boosting researchers' publication record, in the long run it may lead to loss of reading skills (e.g. the researcher cannot distill the most important and relevant information from a paper) and depth of thought, leading to decreased quality of the research output.

3.4. Formulation of research design

AI can help researchers a lot in this direction. Appendix 1 shows the proposed research design (research strategy, population, sampling type, sample size, data collection and analysis methods) for the paper on how cultural differences affect human-robot interactions from the conversations with ChatGPT and Claude. Both AI applications provided suggestions that largely overlap. They are not innovative but relevant and appropriate to the particular topic. The suggestions do not include references but the justification seems appropriate albeit a bit basic. Therefore, researchers can use AI to develop ideas about the specific research design for their studies. The AI's suggestions will be relevant and justified. However, they lack novelty because they are based on something that is already in its database. AI may not propose an innovative methodology that has never been applied to researching a particular topic/problem nor may develop a completely new methodology. Therefore, the methodological contribution of publications in social sciences that implement AI-proposed methodologies may be limited.

3.5. Data collection

AI can be used for data collection in different ways. First, GenAI can help social scientists develop their research instruments or generate visuals (e.g. video, photos) for online experiments. In a recent paper, Hoffmann et al. (2024) developed and validated a scale with statements generated by ChatGPT. Appendix 1 shows the questions that ChatGPT and Claude provided for measuring the human-robot interaction experience. The prompt included the wording that ChatGPT used in its answer to the previous prompt about the formulation of the research design. As evident, the proposed statements

need editing. For example, ChatGPT's answer included a statement ('I have interacted with robots before in my daily life') that cannot be answered on the 5-point level of agreement scale proposed by ChatGPT, while Claude's answer included many statements from the various technology acceptance frameworks rather than focus directly on the interaction experience. Nevertheless, both answers provided good ideas about the statements (e.g. Claude's comfort level statements) that can be further revised before inclusion in a questionnaire. The justification of the statements is also relevant and can help researchers in formulating the methodological arguments in the paper. However, both applications did not provide sources for the statements due to the lack of an explicit request in the prompt. When asked to provide references for the statements, Claude answered that it could not while ChatGPT provided references. A closer look at references showed that the ones checked by the author existed but the statements did not come from them. The references could be used in the justification of research hypotheses and the literature review but they were misleading as sources of the statements. Hence, overreliance on AI to create statements and questions in a questionnaire may backfire. Critical evaluation of the proposed statements is needed to include only those that are relevant. If it is a new construct with new items, GenAI can be actively used and acknowledged but if it is a construct that has been widely researched, it is more convenient to use previous publications as GenAI's answers may be misleading and factually incorrect.

Second, GenAI can serve as a respondent. Since the launch of ChatGPT, thousands of studies, including the current paper, interviewed ChatGPT, Bing, Bard/Gemini or other applications and assessed their answers (e.g. Elmohandes & Marghany, 2024; Soliman et al., 2023). The benefit here is that GenAI can help in new knowledge creation by connecting pieces of knowledge from its database (Williams et al., 2023). GenAI applications can also assume various roles and advise on diverse topics. For example, Almeida and Ivanov (2024) successfully developed a hotel marketing plan based entirely on an interview with ChatGPT. The 'hallucination' problem related to the fabrication of facts remains the main issue. Although it has decreased in new GPT models, it has not disappeared yet, emphasising the need for human researchers' quality control of AI's output and critical thinking.

The third way AI can be utilised in data collection is the generation of synthetic data from silicon samples (see Imschloss et al., 2025; Sarstedt et al., 2024; Viglia et al., 2024; Xiong et al., 2024). In practice, GenAI is used to simulate the behaviour of people with specific characteristics (Park et al., 2024) and respond numerous times to questions posed by the researchers to form a 'silicon' sample which is then analysed. Synthetic data is very useful in testing research instruments because GenAI can provide feedback about the feasibility of the research design, the suitability and wording of questions and answer options and allows researchers to amend the research design and data collection instruments based on GenAI's answers. For the main study, silicon/synthetic data can be valuable for running AI-generated experiments to evaluate the reaction of AI-simulated respondents to different interventions (e.g. price changes or other marketing stimuli). It can also help expand the sample size when it is very difficult to find many human respondents. Here the quality of the training dataset becomes vital because any biases in it (e.g. overrepresentation of some social/respondent groups and under – or misrepresentation of others) may result in creating AI output that significantly deviates

from what human respondents would provide. Nevertheless, regardless of how realistic such AI-generated experiments and synthetic data are, they are only a substitute for and a surrogate of real-life data and do not reflect entirely the behaviour and responses of humans. Therefore, synthetic data from simulated AI respondents is of lower real-life relevance and validity compared to data on the same topic from human respondents. Hence, synthetic data must be benchmarked against data from human respondents for validity and reliability (Sarstedt et al., 2024).

Finally, chatbots may assist in data collection by distributing a questionnaire, either as an external link or embedded within the chatbot. The interactivity of chatbots, especially when they are connected to LLMs, may contribute to a higher response rate, and the collection of higher-quality answers to more open-ended questions compared to traditional web-based questionnaires (Xiao et al., 2020). Therefore, chatbots can be useful in the simultaneous collection of both quantitative and qualitative data. Additionally, chatbots may help reach respondents from groups present on social media platforms that might be otherwise underrepresented in web-based or paper surveys. From an ethical perspective, the use of a chatbot for data collection requires that the chatbot explicitly identifies itself as a chatbot before the start of the questionnaire to avoid potential confusion among respondents. However, Zarouali et al. (2024) found that chatbots performed worse in terms of respondent enjoyment, perceived privacy and data quality compared to web surveys. Moreover, social media platforms might use the conversations between the chatbots and respondents for user profiling. The lack of complete control over the data and the data collection process in chatbot-based surveys is a significant ethical issue that might limit their practical application for research purposes.

3.6. Data analysis

Although AI has long been used for data analysis in social sciences (Christou, 2025; Lee et al., 2022), its application was limited due to the high technical and computer skills required by researchers. GenAI decreased drastically the skills requirements and democratised AI, stimulating its widespread use for data analysis. The advances of GenAI allow its implementation for quantitative and qualitative data analysis (Perkins & Roe, 2024). As quantitative analysis is quite straightforward, Appendix 1 shows only one of the ways GenAI could be employed for qualitative data analysis – sentiment analysis of customer reviews. The advantage of GenAI is that it can provide a structured response and justification of its evaluation. However, as evident from Appendix 1, two GenAI applications can elicit quite different outcomes. Using the same prompts and text of the customer review, ChatGPT evaluated it as positive while Claude as neutral. Although both logically justified their evaluations, the difference in the outcome is substantial. Hence, from an ethical perspective, researchers should not rely on (one) AI tool only for textual analysis but combine different tools and manually check and validate a sample of GenAI's output (Ivanov et al., 2025). Overreliance on AI for data analysis may lead to misleading results and irrelevant conclusions.

Transparency, traceability, and explainability are major concerns in AI analysis. Researchers need to know how exactly a specific decision (evaluation, outcome) of an AI tool was obtained. Transparency and traceability require that researchers provide

the prompts for the AI-generated content (Kendall, 2024), screenshots and the links to the conversation with the AI tool as done in Appendix 1. This allows other researchers to check the AI output and rerun the prompt with the same or another GenAI application to validate the results. Such an approach also protects the researchers as they can prove how the AI output was obtained. For non-GenAI models (such as models for quantitative data analysis and forecasting), the training datasets and model specifications can also be made available either publicly or provided by the authors upon reasonable request. However, as already discussed in Section 2.3 the principles of transparency and traceability may conflict the intellectual property rights of AI algorithm developers. Therefore, the provision of the algorithms for audit by respective institutions (e.g. Ethics committees) is the practically reasonable solution for the moment.

3.7. Writing/manuscript preparation

Writing a manuscript is a major task that many researchers would love to outsource to an AI application. In practice, every response to a prompt that GenAI provides can be copied, pasted, edited, and potentially used in a scholarly paper; thus, Appendix 1 does not include a prompt and respective responses from GenAI. The use of GenAI in writing the paper goes beyond the idea generation, translation of text and summary of a paper's findings. It is related to directly generating text that is used in a manuscript with minimal editing. While GenAI applications improve the efficiency of researchers in writing manuscripts (Madanchian & Taherdoost, 2025), they reshape academic publishing beyond the sheer efficiency (Van Quaquebeke et al., 2025) and raise numerous ethical questions.

The most important unethical practice is the fabrication of parts of or whole academic papers (Chauhan & Currie, 2024; Nazarovets, 2024). The democratisation of GenAI decreased the costs in terms of money, time, and necessary skills for creating fake manuscripts with fabricated data or results based on non-existent (empirical) studies. Previous studies have shown that GenAI can create very persuasive content (Salvi et al., 2025). GenAI can generate convincingly sounding but entirely fake texts, datasets, tables, figures, and references. Often such fabricated papers are developed by paper mill companies that sell papers to authors desperate for publications. When published, fabricated papers tarnish and skew the scholarly record and misinform managerial and policy decisions because they report findings of studies that never took place. These fabricated studies furthermore decrease the effectiveness of GenAI because they may be used to train its new versions which will be based on biased dataset that includes non-existent studies. When identified, fabricated papers are retracted and damage the reputation of the authors and the journals they are published.

Another ethical issue relates to the authorship of the manuscript. Currently, GenAI applications do not meet the requirements for authorship because unlike humans they cannot take responsibility for the text they generate. But if the majority of the text was generated by AI, do the human authors have sufficient intellectual input in the manuscript to justify authorship? For the moment, the answer to this question is 'Yes' because humans write the prompts but there are more than fifty shades of grey in the answer depending on the degree of human involvement in the final version of the

text. Responsible AI use requires human-AI co-creation of the text (see, for example, Tarbit et al., 2023) rather than a simple copy/paste of an AI-generated text. Through a series of prompts, AI-generated texts, summaries and analyses, and heavy human editing of the text, the final output of AI-assisted writing is a manuscript where AI-generated texts are skilfully interwoven and merged with human-written texts without linguistic differences between them. The intellectual input of the author changes from writing the text of the manuscript to writing the prompts and editing the texts GenAI wrote.

A third ethical issue relates to plagiarism which may arise when GenAI does not acknowledge the sources it used to generate the text, cites wrong sources or generates strikingly similar texts to different users. Moreover, the text can be factually incorrect and invent facts. Publishers perceive GenAI as a tool and their guidelines explicitly state that the human authors take full responsibility for any text or visual generated by AI. In case of plagiarism and factual incorrectness, the authors cannot blame the AI but must take responsibility for the unfair practice and face the consequences.

Finally, GenAI applications use specific words (e.g. 'underscore', 'elucidate', 'illuminate', 'foster', etc.), syntax (e.g. 'While [gerund/noun + verb] ... , [gerund/noun + verb] ... '), or response structure (e.g. general 1-2-sentence Introduction, bullet-pointed answer and a conclusion paragraph starting with 'Overall' – see ChatGPT's answer to the data analysis prompt in Appendix 1) that are repetitive and make the text easily recognisable as AI-generated. Additionally, GenAI applications often use synonyms for terms when they should not due to the different meaning of the synonyms in the research context. The substitution of words by synonyms might be initiated by the human authors too to avoid potential plagiarism detection but such practice often leads to tortured phrases that deviate from the established terms such as 'enormous information' instead of 'big data', and can signal fabricated papers (Else, 2021). Unnecessarily complicated and flamboyant ways of expression are also characteristic of many GenAI answers and are inconsistent with good academic writing. If AI-generated texts are to be used in a manuscript, they need to be edited for correctness of terminology and proper wording.

3.8. Manuscript evaluation (review and editing)

AI can be effectively used to evaluate manuscripts. Some applications (e.g. <https://statcheck.io/>) allow users to check the correct reporting of some statistical analyses while GenAI applications have more versatile capabilities. Appendix 1 shows ChatGPT's and Claude's assessment of a manuscript written by the author of this paper, reviewed and rejected by a journal two decades ago and never resubmitted to another journal (hence, there are no copyright issues). GenAI's comments and recommendations were very relevant, detailed, focused, and reflected the actual quality of the manuscript. This was done in minutes rather than hours of work for the editor and the reviewers. Hence, GenAI can save a lot of time for them. However, currently, publishers do not allow reviewers and editors to use GenAI for the evaluation of manuscripts due to various reasons (see for example Elsevier's policy here – <https://www.elsevier.com/about/policies-and-standards/the-use-of-generative-ai-and-ai-assisted-technologies-in-the-review-process>). First, until a manuscript is accepted and copyrights are transferred to

the publisher (if the paper is published in a pay-to-read mode), it belongs to its authors. Using AI to evaluate the manuscript requires the explicit permission of the authors. Currently, submission platforms do not request such explicit permission from the authors. Second, by uploading the file on an LLM-based chatbot, the manuscript can become part of its database. Therefore, it can generate texts that are similar to the one it evaluated. In practical terms, if the reviewers of manuscript A used GenAI to assess it, this application may generate texts similar to the text in manuscript A which other users can utilise in manuscript B and acknowledge accordingly. When both manuscripts are published, there may be an overlap between the texts in them and a potential accusation of plagiarism. The authors of both manuscripts might have acted ethically but their manuscripts may have overlapping texts due to reviewers' or editors' actions. Third, LLMs have not reached the level of sophistication to replace the expert judgement of experienced reviewers and editors. The manuscript used to test the capabilities of the two AI apps was correctly rejected by the three reviewers, the handling editor and the Editor-in-chief while both apps recommended a revision and provided the necessary comments. The responsible use of AI requires reviewers and editors to not upload the reviewed manuscript in a GenAI application to protect the interests of the authors despite the potential efficiency gains. If GenAI is to be used, authors need to be aware of its application in the editorial and review process. The file should not remain in the application's memory to be used for generating other texts. Finally, from operations management perspective, the reviewers were invited to review the manuscript based on their subject knowledge and methodological expertise, not their skills to use GenAI applications. If editors and publishers wanted the manuscript be evaluated by AI they would have done it by themselves. In the future, AI will be more actively used to evaluate manuscripts due to the growing number of submissions and the lack of sufficient competent reviewers ready to volunteer. This means that researchers would need to write not only for humans (editors, reviewers, readers) but for AI reviewers as well. This may change the writing style in papers. Some authors have already recognised this fact and have infused in their manuscripts hidden prompts to AI reviewers to provide positive reviews only (Sugiyama & Eguchi, 2025).

Publishers can arm editors with AI tools (e.g. Proofing.com, Imagetwin.ai) to catch manipulated images, inconsistent texts, image duplications, etc. Other tools identify AI-generated texts (e.g. ZeroGPT, Turnitin) or check references (e.g. Scite.ai) (Andrade-Hidalgo et al., 2024). In doing so, some of the unethical uses of AI by authors and reviewers can be caught during the review process before a manuscript is published, decreasing the probability of a potential retraction if the paper is published with unethical content. However, the author's experience shows that the results of these applications are still unreliable as they give strikingly different results for the same text (no AI-generated text by one application, 100% AI-generated by another one), mark human-written text as AI-written, and mark different parts of a text as potentially AI-generated, decreasing the trust in their output. This may lead to unsubstantiated accusations of authors of AI plagiarism, paper fabrication or other unethical practices (Giray, 2024). Hence, human judgment is necessary rather than overreliance on AI tools.

Table 1 summarises the guidelines for responsible AI use outlined above. They are not written in stone and will evolve in line with technology's capabilities.

Table 1. Guidelines for responsible AI use in social science research.

Research stage	DOs	DON'Ts
Ideation	• Generate ideas about research projects • Use a series of prompts to direct AI to elicit more innovative answers	• Overreliance on AI as ideas may be descriptive, trivial and without theoretical, methodological or practical contribution
Problem definition	• Draft and edit the manuscript's aim and objectives • Generate ideas about the problem justification and the research rationale	• Copying and pasting the generated texts without critical evaluation as the aim, objectives and rationale might be descriptive, incoherent and/or irrelevant
Review of previous studies	• Summarise papers and identify key points in them • Compare key points in publications (methodologies, findings) • Check every reference cited by the AI application • Develop customised GPT with selected publications	• AI should not replace the actual reading of manuscripts by the researcher • Rely on AI-generated reviews as they tend to be descriptive, lack critical evaluation of the literature, and may omit important papers or key points in them
Formulation of research design	• Generate ideas about the research design • Compare alternative research designs • Justification of the research design	• Rely blindly on the research design proposed by the GenAI without appraisal of its relevance and applicability.
Data collection	• Clearly indicate synthetic data and conclusions based on it • Generate ideas about constructs and statements to measure them • Draft and edit questions for a questionnaire or interview • Test the data collection instrument (e.g. a questionnaire) and receive feedback from GenAI about it	• Use synthetic (silicon) samples when human respondents could be approached.
Data analysis	• Use AI to analyse data • Keep a record of the input (e.g. prompts) and output (e.g. generated text) of the AI tools for transparency and traceability of the analysis. Provide a link to the conversation with the GenAI tool. • Use different AI applications to cross-validate results	• Generalise conclusions from synthetic samples for the human research population • Consider that AI's analysis of (qualitative) data is authoritative
Writing/Manuscript preparation	• Heavy editing to 'humanise' the AI-generated text • Check the text for proper use of terminology and factual correctness • Proofreading and rewording • Open access to data, prompts, AI output and AI models in an Appendix or a repository for audit purposes • Be careful who you work with and show zero tolerance towards unethical practices	• Generate, copy and paste the text of the manuscript from GenAI applications • Lack of transparency • Fabricate data, results, references, facts, visuals • Add a GenAI app as a co-author
Manuscript evaluation (review and editing)	Editors and reviewers: • Adhere to the publisher's guidelines for manuscript evaluation and review • Use digital tools to identify duplicated images and texts, check the correctness of statistical analyses, validate references, etc.	Editors and reviewers: • Upload the manuscript in an online AI application (e.g. ChatGPT) for automatic evaluation without the explicit consent of the authors Authors: • Blame AI for unethical practices

(Continued)

Table 1. Continued.

Research stage	DOs	DON'Ts
	Authors: • Take full responsibility for the manuscript content, including the part generated by AI • Acknowledge the tasks AI was used for	

4. Conclusion

4.1. Contribution

The paper has several contributions. First, it outlines the principles of responsible AI use in social science research, namely: transparency, traceability/auditability, explainability, fairness, privacy and data protection, respect of all research stakeholders and awareness of broader impacts of AI use in research, human oversight and quality control, and accountability. Adherence to these principles ensures that AI-assisted research will be implemented ethically. Second, the paper operationalises the principles into specific actions to be followed by research stakeholders in each step of the research process – from ideation to manuscript review. Finally, the paper practically illustrates the capabilities, strengths and weaknesses of AI in each step of the research process by utilising two popular GenAI applications.

4.2. Implications for authors

The temptation to use AI to gain a competitive advantage in the ‘publish or perish’ game is substantial. As the paper showed, researchers can use AI in various irresponsible ways to speed up the production of manuscripts and increase publication count. Although winning in the short term, such a strategy would backfire in the long term. Ultimately, authors are responsible for all the content that AI generates and AI tools cannot be used as scapegoats for plagiarism, factual incorrectness, wrong terminology, or other unethical practices – all these remain the full responsibility of their human users as the accountability principle requires. Although a recent study showed that AI agents could successfully implement most tasks in the research process (Schmidgall et al., 2025), overreliance on AI for literature reviews and manuscript text generation means insufficient reading leading to underreliance on and underdevelopment of the author’s cognitive and critical thinking skills. Biased datasets to train AI models will produce biased and invalid research results. In this regard, authors should be very careful whom they cooperate with because their co-authors might not be ethical, use AI unethically, and lack proper research skills. Researchers may follow all good ethical practices but still face ethical issues in their research projects and publications due to the unethical behaviour of their co-authors.

4.3. Implications for PhD students and early career researchers and PhD programmes

Universities need to develop internal guidelines and provide proper training to early career researchers, PhD students and their supervisors on the responsible use of AI in

research. Training for students from all levels can be provided within the *Research methods* or similar modules. Doctoral programmes may include a dedicated module on *AI in Research* where its application is discussed in greater detail. On a conceptual level, students need to understand the AI governance frameworks (e.g. EU AI Act), the social and economic impacts of AI in research and its bright and dark sides. This will help them understand the reasons behind the need to use AI responsibly, the principle of responsible AI use in research and the broader impacts of AI in research. On a practical level, students need to learn how to develop AI models and appropriate prompts for effective and efficient use of AI tools, what to use AI for and what to avoid, and how to ensure transparency, traceability, fairness, explainability and privacy of the results, and the methodological procedures in their research. This will help them learn the necessary actions to ensure they use AI ethically. Supervisors need to receive periodic training to refresh and upgrade their skills and to mentor properly their (PhD) students. The guidelines and training will evolve with the advancement of AI's capabilities.

4.4. Implications for publishers

Fabricating references has negative consequences beyond the manuscript and its authors. It pollutes the scholarly record with references to non-existent studies. Moreover, the assumed authors of the fabricated references might face ethical issues. For example, as of July 2025 the author of this article found references to eleven non-existent papers for which he was the assumed co-author. The papers were on his usual topics, most co-authors of the fabricated papers had co-authored actual papers with him, and the journals they were supposed to have been published in existed and were relevant. However, the papers did not exist. One can only wonder how many similarly fabricated references exist in the scholarly record and how they will shape policy decisions. Publishers need to take serious action by validating each reference. The users of some manuscript management platforms already have the option to validate references based on DOI links and indexing in Scopus/Web of Science, but many publishers do not have the technical capability or financial resources to do it. A recent initiative by the *European Journal of Tourism Research* and *ROBONOMICS: The Journal of the Automated Economy* requires all new submissions to provide DOI or other links to the actual publications cited in the manuscripts, including books and conference proceedings, while publications that cannot be traced in the Internet and validated should not be included in the reference list. While such actions may not entirely eliminate fabricated references because generative AI can fabricate DOI and web links as well, they will decrease them.

Publishers struggle to find competent reviewers due to their workload and the virtually free labour of the reviewers. A solution to this review crisis in publishing could be the active use of AI (see also Bagenal, 2024). For example, the review of a manuscript might be divided into two stages. Initially, a manuscript is checked by an AI and only the 30-40% of submissions that pass its check go to the next stage for the human review. The AI review may focus on the geographic and thematic scope of the paper and basic issues in the manuscript which the LLMs can identify and, if applicable, the correctness of the statistical analysis and its reporting (see Appendix 1). The AI can prepare draft comments for the Editors who can revise them as appropriate. In that way, the

review process will speed up and the workload of reviewers and editors will be more manageable. Additionally, AI might be used to detect fake paper reviews, similar to the detection of fake customer reviews of restaurants or hotels (Lee et al., 2022).

4.5. Implications for funders

Similar to higher education institutions, research funders need to develop guidelines for the responsible use of AI in the projects they finance to stimulate transparency, traceability, fairness, privacy, explainability and accountability in AI use. They can organise or support (online) training programmes for responsible AI use in research and require adherence to specific AI governance frameworks. In addition to the data management plan and research ethics, project applications need to include a section elaborating how AI will be used, how it will affect participants, researchers, their institutions, society as a whole, and other relevant stakeholders, and outline the respective AI risk mitigation strategies. Funders might mandate the open-access publication of methodologies, datasets, and AI models in repositories to facilitate their audit and ensure replicability and transparency of results and methodological procedures.

4.6. Implications for policymakers

The unethical use of AI in research may have significant negative societal impacts. Biased datasets and fabricated studies can misinform policy decisions and decrease public's trust in science. To mitigate the potential negative societal impacts and promote responsible AI use in research, policymakers need to develop AI governance frameworks, based on global/internationally recognised frameworks such as the EU's AI Act, that elaborate the main principles of responsible AI use in research which higher education institutions, research organisations and funders can operationalise into their own specific rules and regulations. They can mandate research institutions to establish AI oversight boards and provide AI training to their staff and students, require the open-access publication of AI datasets and models, and provide the technical infrastructure (i.e. repository) for this. Policymakers may request periodic reports from research organisations on responsible AI use in them which can also be part of the accreditation process.

4.7. Future prospects

Researchers have already made significant steps in the brave new world of human-AI collaboration in research. Using AI in social science research was exceptional until a few years ago, it is now mainstream and may become compulsory in the future. AI-related skills will become part of the requirements for PhD students and researchers and embedded in the study curricula. Journals will expect more complex research designs and analyses forcing researchers to utilise AI for data collection and analysis. Universities and research organisations will increase their expectations about the publication output pushing researchers to use AI for writing manuscripts. The recent initiative of *ROBONOMICS: The Journal of the Automated Economy* for AI-co-created articles may become the new norm while publications that do not utilise AI in any form may be rare. Publishers may even start accepting manuscripts entirely written by AI similar to the first AI-authored book published by

Springer in 2019 (see Writer, 2019). Editors and reviewers of manuscripts and project applications/proposals will have the support of AI research assistants who will make initial editorial decisions, outline potential strengths and weaknesses of the evaluated manuscripts/project proposals and draft decision letters. This will force researchers to change their academic writing style to adjust it to the AI filters publishers and funders will likely introduce as a solution to the reviewer crisis caused by the insufficient number of reviewers to face the number of submitted manuscripts/project proposals. Academic papers may become more transparent, including raw data and AI code that can be easily audited by publishers' AI research assistants. The style of writing may be less convoluted and more straightforward which will benefit non-native speakers and will increase the readability of papers and their usefulness for managers and policy-makers. Ultimately, these actions may increase the general public's trust in science. At the same time, the proliferation of synthetic (AI-generated) content may decrease the efficiency and relevance of some research methods. According to the Dead Internet Theory (Walter, 2024), much of the content and traffic on the internet is already generated by AI. The wider adoption of GenAI would increase this share even more. This means that it will become increasingly difficult to conduct studies based on analysis of online content because the conclusions from it might be misleading as most of the content will be created by AI and will not reflect humans' actual opinions, ideas and aspirations.

Fascinating as it may be, this brave new world of human-AI collaboration in research is an ethics minefield whose cleaning requires the concerted efforts of all research stakeholders. Publishers may need to relinquish some of their copyrights and allow for AI agents owned by other publishers/organisations to freely use their content so their own AI agents can use the content owned by other publishers/organisations as well. Pooling academic content will improve the quality and decrease the biases in the training datasets of research-focused AI chatbots. Funders, publishers, higher education and research organisations, IT companies, and publishers need to work together to harmonise their ethical guidelines of AI use which then should be adopted and rigorously applied by them. AI algorithms used in research should be made available for audit by the respective authorities. To protect their developers' intellectual property rights and financial investments, the AI algorithms should not always be publicly available but should be provided for inspection in case there are serious and justified concerns about the ethical aspects of their output. All these coordinated actions will require legal changes in intellectual property rights on global level, push by publisher associations, funders, universities, and legal authorities, and changes in the territorial way of thinking ('this is ours, not theirs') of some people sitting in corporate boards; hence, they may face some resistance. The exponential development of AI's capabilities and its growing use in research will, however, force research stakeholders to take the necessary actions.

4.8. Limitations and future research

The main limitation of this paper is its conceptual nature. Future research needs to investigate empirically the adoption of AI at the various stages of the research process and the factors that drive it. Studies need to shed light on researchers' inclination to misuse AI and the respective influencing factors to develop effective mitigating strategies. Additionally, research may assess the efficiency of responsible AI guidelines and training of universities,

research organisations, funders and publishers. Research can focus on the publishers', editors', reviewers' and authors' readiness to accept and use AI in the editorial and review process of manuscripts. A Delphi study may be implemented with leading social scientists to distil the good practices in responsible AI use in all stages of the research process and those practices that need to be avoided, building on the current list in Table 1. From a methodological perspective, research needs to be directed towards the improvement of existing and development of new research methodologies based on AI. Finally, research can investigate whether AI delivers to the hype and really improves the quality and efficiency of the research process and the academic publications or whether its contribution is overestimated.

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No potential conflict of interest was reported by the author(s).

Data availability statement

Data from generative AI is included in Appendix 1. There is no data from human respondents.

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